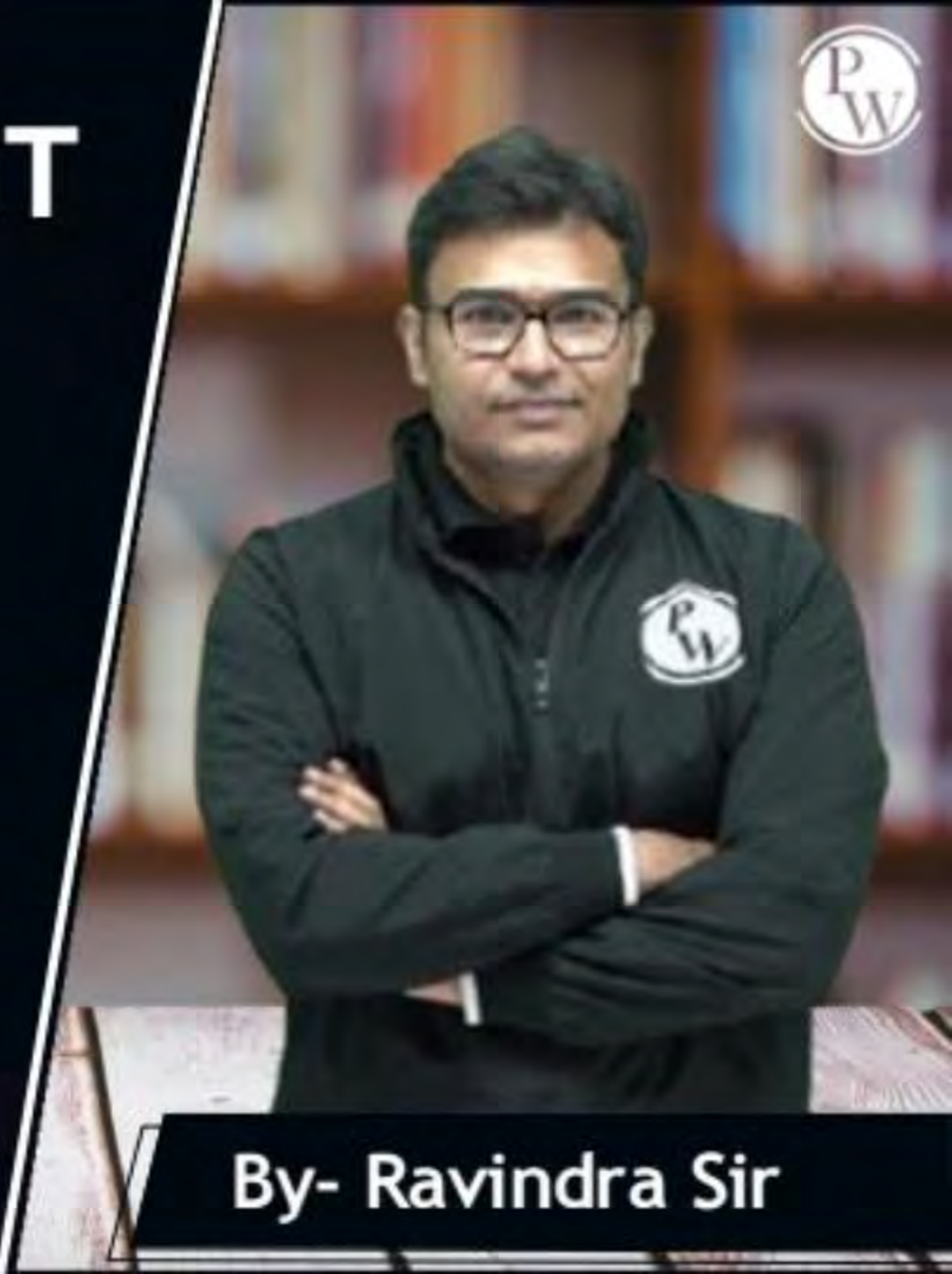
Computer Science & IT

COMPILER DESIGN
(CD)

Intermediate Code Generation

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

ICG

Topics to be Covered



- Topic
- Topic
- Topic

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

while E do S



L: if (E==0) goto L_1

S ✓

Goto L ✓

L_1:

L: if (E) goto L_1

goto Last

L_1: S

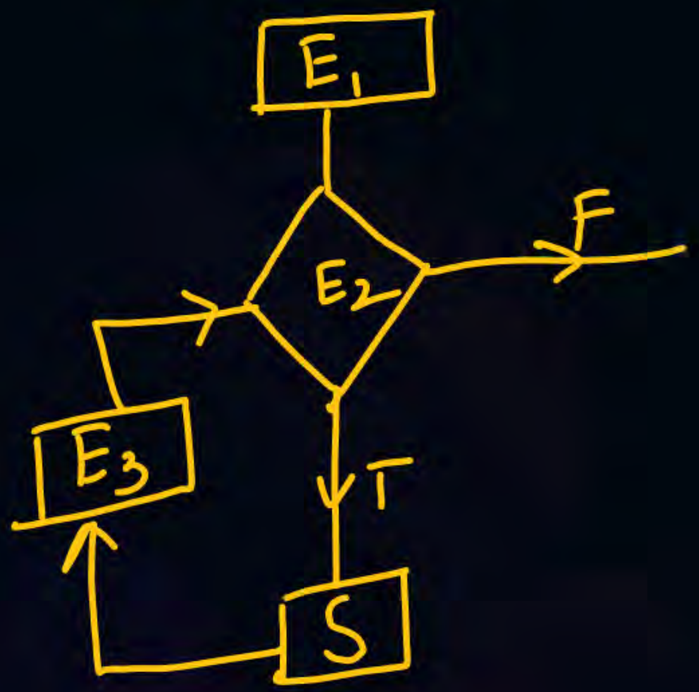
goto L

Last:

while (a < b) do
    x = y + z

L: if a < b goto L1
   goto Last
L1: t = y + z
   x = t
   goto L
Last:

for (E_1 ; E_2 ; E_3)
S



```
for(i=0; i<10; i++)
    a = b+c;
```

```
    i=0
L:  if (i<10) goto L1
    goto Last
L1: t1=b+c;
    a=t1
    t2=i+1
    i=t2
    goto L
Last:
```

```
switch(i+j)
{ case(1): a=b+c;
           break;
  can(2):  P=Q+R;
           break;

  default: x=y+z;
           break;

}
```

```
t=i+j
goto test

L1: t1=a+b
    a=t1
    goto last

L2: t2=Q+R
    P=t2
    goto last

test: if(t==1) goto L1
      if(t==2) goto L2
      goto L3

L3: t3=y+z;
    x=t3;
    goto last

Last:
```

Which of the following is an intermediate representation

a) Postfix notation

b) 3 add code

c) quadruples

d) all the above

Three add code involves

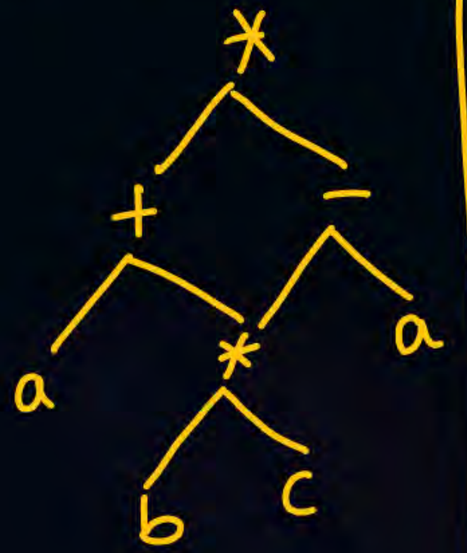
a) exactly 3 addresses

b) at most 3 addresses

c) no unary operator

d) none of them

The equivalent expression fd the following DAG?



a) $((a+b)*c)*(b*(c-a))$

b) $a+(b*c)-a$

c) $(a+(b*c))*((b*c)-a)$

d) $a*(a+b*c)-a$

THANK - YOU